

Concept: Eight supplementary demand/band calibration tasks

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Eight supplementary demand/band calibration tasks:

Supported practice

AUTHORED practice

Subject: Mathematics

Topic: Algebraic decisions and domain boundaries

Concept: Classify after putting the equation in standard form

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Classify after putting the equation in standard form

Question type: Constructed response

Provenance: Authored practice

Classify after putting the equation in standard form

Over the rationals, classify $3x - 4 = 1 - 2x$ as having no solution, one solution, or every rational as a solution. Use the criterion $Ax = B$: $A \neq 0$ gives one rational solution, $A = B = 0$ gives every rational, and $A = 0$ with $B \neq 0$ gives none. Do not report the value of x .

1. Keep the question about the size of the solution set, rather than a numerical root.

Conditions

- Original calibration practice; independent educational review pending.

Requested output: category of solution set

$$3x - 4 = 1 - 2x$$

No solution | One solution | Every rational

Givens and requested output; Construction framework

AUTHORED practice

Subject: Mathematics

Topic: Algebraic decisions and domain boundaries

Concept: Classify linked parameter exceptions

Concept difficulty: Medium concept

Question difficulty: Multi-step reasoning

Family: Classify linked parameter exceptions

Question type: Constructed response

Provenance: Authored practice

Classify linked parameter exceptions

Let k be rational and x rational. For each k , classify each of these equations as having none, one, or every rational solution: E1: $2(x-3)=2x+k$; E2:

1. Keep the two equations separate; k is a fixed parameter when you classify solutions in x .

$(k-2)x=4-2k$. Use the $Ax=B$ criteria; a numerical solution is not requested.

Conditions

- Original calibration practice; independent educational review pending.

k is fixed; x ranges over rationals

E1: $2(x-3) = 2x+k$

E2: $(k-2)x = 4-2k$

Stage 1 of 2

Givens and requested output; Construction framework

AUTHORED practice

Subject: Mathematics

Topic: Algebraic decisions and domain boundaries

Concept: Classify with a coupled domain boundary

Concept difficulty: Medium concept

Question difficulty: High-transfer / exception-sensitive

Family: Classify with a coupled domain boundary

Question type: Constructed response

Provenance: Authored practice

Classify with a coupled domain boundary

Extension beyond routine linear equations. k is real, x must be rational. Classify the simultaneous system by k as INVALID PARAMETER, NO RATIONAL SOLUTION, ONE RATIONAL SOLUTION, or EVERY RATIONAL SOLUTION: $(k-1)(k-2)x=k-1$ and $(x+1)/(x-1)=k/(k-2)$. An equation with a zero denominator is invalid; a solution must satisfy both original equations and $x \neq 1$. Report

1. Distinguish an undefined equation from an equation that is defined but has no rational solution.

categories and the parameter cases, not a general numerical solution method.

Conditions

- Original calibration practice; independent educational review pending.
- Advanced parameter/domain extension; not an ordinary Grade 9 retrieval question.

k real; x rational; both equations required

$$(k-1)(k-2)x = k-1$$

$$(x+1)/(x-1) = k/(k-2); \text{ denominators nonzero}$$

Stage 1 of 2

Givens and requested output; Construction framework

AUTHORED practice

Subject: Mathematics

Topic: Algebraic decisions and domain boundaries

Concept: Execute an inverse chain across a table and a process graph

Concept difficulty: Medium concept

Question difficulty: Multi-step reasoning

Family: Execute an inverse chain across a table and a process graph

Question type: Constructed response

Provenance: Authored practice

Execute an inverse chain across a table and a process graph

A measuring program uses numerical readings in centimetres: $X \rightarrow A = (X - 1)/2 \rightarrow B = 3A + 4 \rightarrow C = (B + 2)/5$. The observation table gives $C = 22$ millimetres. The inverse procedure is prescribed: convert the observation to centimetres, undo the C operation, then B, then A. Execute it to recover

1. Keep the observation unit separate from the program unit.

X and verify every forward stage. Treat the displayed readings as exact.

Conditions

- Original calibration practice; independent educational review pending.

Observation table: $C = 22$ mm; program uses cm

$X \rightarrow$ subtract 1, divide 2 $\rightarrow A$

$A \rightarrow$ multiply 3, add 4 $\rightarrow B \rightarrow$ add 2, divide 5 $\rightarrow C$

Stage 1 of 2

Givens and requested output; Construction framework

AUTHORED practice

Subject: Mathematics

Topic: Algebraic decisions and domain boundaries

Concept: Choose the governing equation from its conditions

Concept difficulty: Medium concept

Question difficulty: Direct application

Family: Choose the governing equation from its conditions

Question type: Constructed response

Provenance: Authored practice

Choose the governing equation from its conditions

A linear calibration is stated as $Ax=B$ with rational A and B , and A is known to be nonzero. Choose the governing rule for recovering x : R1, $x=B/A$; R2, infer every rational is a solution from $0=0$. Only choose the applicable rule; do not compute a value.

1. Identify the condition on the coefficient before choosing a rule.

Conditions

- Original calibration practice; independent educational review pending.

Given: $Ax = B$; $A \neq 0$

Candidate R1: divide by A

Candidate R2: zero-coefficient identity

Stage 1 of 2

Givens and requested output; Construction framework

AUTHORED practice

Subject: Mathematics

Topic: Algebraic decisions and domain boundaries

Concept: Translate an ordered pair without changing meaning

Concept difficulty: Medium concept

Question difficulty: Direct application

Family: Translate an ordered pair without changing meaning

Question type: Constructed response

Provenance: Authored practice

Translate an ordered pair without changing meaning

The ordered-pair table has columns x and y and one row $x=2, y=3$. Write that row as a point in Cartesian ordered-pair notation. The coordinate convention is horizontal x first and vertical y second; no equation solving is requested.

1. Keep each value attached to its column heading.

Conditions

- Original calibration practice; independent educational review pending.

Table heading: $x \mid y$

Row: $2 \mid 3$

Coordinate template: (horizontal, vertical)

Stage 1 of 2

Givens and requested output; Construction framework

AUTHORED practice

Subject: Mathematics

Topic: Algebraic decisions and domain boundaries

Concept: Order two dependent inverse steps

Concept difficulty: Medium concept

Question difficulty: Direct application

Family: Order two dependent inverse steps

Question type: Constructed response

Provenance: Authored practice

Order two dependent inverse steps

A starting number is first increased by 3 and then doubled, producing 14. You know the inverse components: subtract 3, and halve. Coordinate them into the correct reverse order. State which

1. The final output comes from the last forward operation.

component must come first and why; do not calculate the starting number.

Conditions

- Original calibration practice; independent educational review pending.

Start → add 3 → double → 14

Inverse components: halve | subtract 3

Required: dependency-respecting order

Stage 1 of 2

Givens and requested output; Construction framework

AUTHORED practice

Subject: Mathematics

Topic: Algebraic decisions and domain boundaries

Concept: Give the warrant for a reversible division

Concept difficulty: Medium concept

Question difficulty: Direct application

Family: Give the warrant for a reversible division

Question type: Constructed response

Provenance: Authored practice

Give the warrant for a reversible division

Over the reals, an author concludes $x=4$ from $2x=8$ by dividing both sides by 2. Supply one warrant explaining why the conclusion follows and why the same justification would fail for division by zero. No further equation solving is requested.

Conditions

- Original calibration practice; independent educational review pending.

1. The result is already stated; the question asks what licenses the step.

Premise: $2x = 8$

Claim: $x = 4$

Requested: a warrant for the division step

Stage 1 of 2

Givens and requested output; Construction framework

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